

Cell and Cancer Biology

Hallmarks of Cancer

Integrated Biomedical and Clinical Sciences OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Objectives

1. Describe the processes in cell life and death.
2. Compare and contrast apoptosis and necrosis.
3. Compare and contrast benign and malignant tumors.
4. Describe hallmarks of cancer.
5. Discuss prevention and treatment strategies.

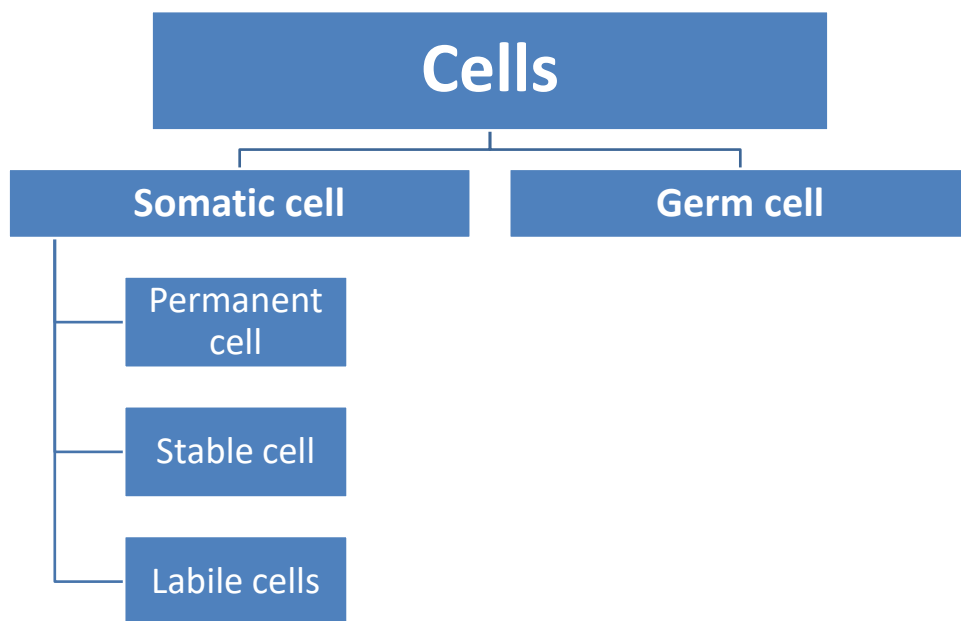
Outline

- Cell Cycle.
- Response to cell injury
- Global burden of Cancer.
- Cancer genes.
- Carcinogenesis.
- Hallmarks of Cancer.
- Prevention and treatment strategies.

Cell Life and Death

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Control for Cell Growth/Division

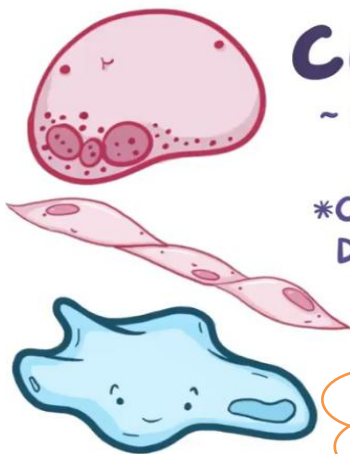
For a cell to proliferate/divide, it receives an extracellular signal from other cells. Those signaling molecules can be:

- (1) mitogens:** Molecules that activates cell proliferation/division via stimulating DNA synthesis. Cells deprived from mitogen signal can enter a non-proliferating state.
- (2) Growth factors:** stimulate protein synthesis of proteins and inhibit protein degradation, for example NGF, IL-2.
- 3) Survival factors** suppress apoptosis (programmed cell death), for example members of Bcl-2. Overexpression associated with tumors.

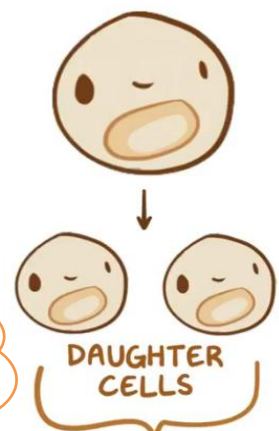
CELL CYCLE

~ SOMATIC CELL SEQUENCE
(FORMATION → DIVISION)

*CYCLE VARIES IN LENGTH
DEPENDING ON CELL TYPE



Epidermal cells complete a cell cycle <1 day while liver cells can take years for one cell cycle.



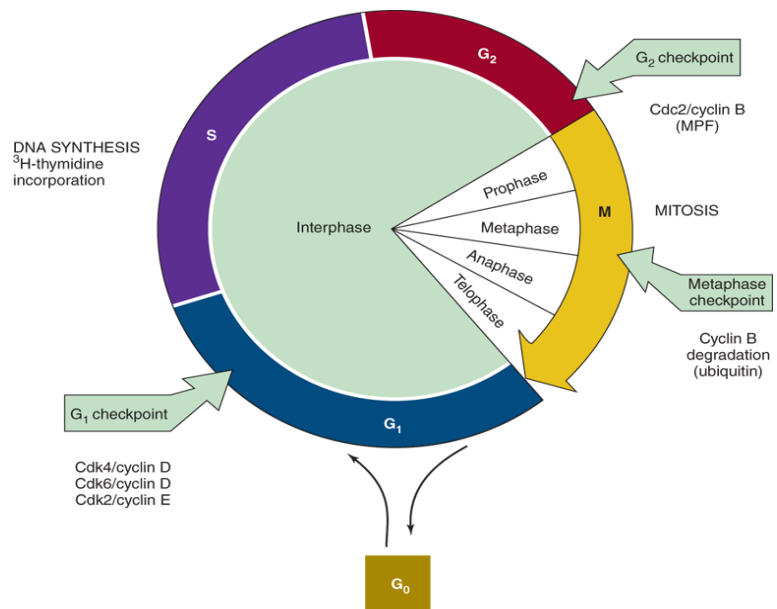
DAUGHTER
CELLS
SKIN CELLS

OSMOSIS.org

A) Interphase

- 1- G₁ phase
- 2- S phase
- 3- G₂ phase

B) Mitosis



Source: Douglas F., Paulsen:
Histology & Cell Biology: A USMLE Step 1 Review, 6e
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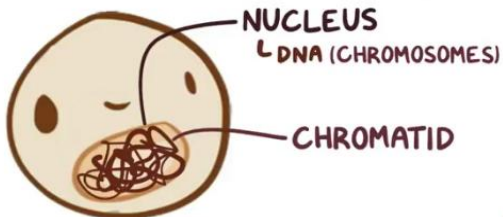
Interphase “G1”

G1 (GAP OR GROWTH 1)

~LONGEST PHASE

*CELL GROWS

*ORGANELLES SYNTHESIZE
PROTEINS & PRODUCE ENERGY



Length and transition rate varies
among different cell types

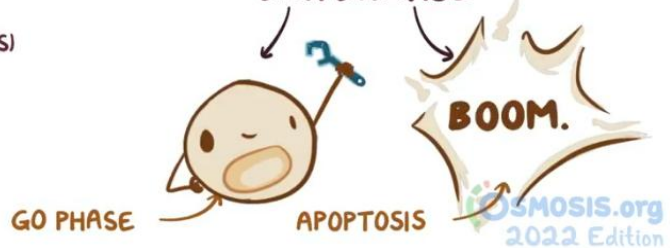
G1 CHECKPOINT

CHECKS FOR:

*DAMAGED DNA

*RIGHT PROTEINS

DNA DAMAGE

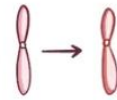


From G1, cells can go to S or G0 phase.

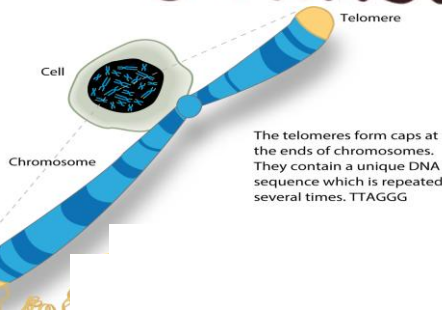
- In G₀ phase, some cells become terminally differentiated and are unable to reenter the cycle (like, muscle, nerve cells). Other cells take a pause and reenter the cell cycle in response to growth factors during or after injury (most common, hepatocytes and fibroblasts).

S PHASE ~SYNTHESIS

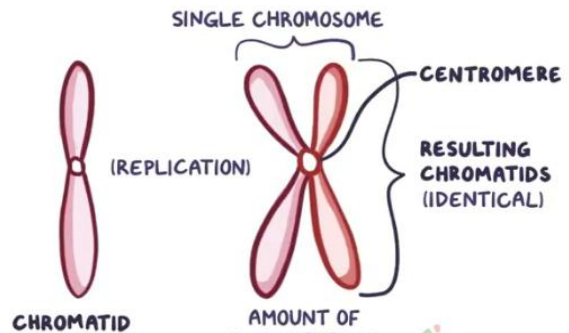
*DNA IS REPLICATED



← (RECEIVE IDENTICAL COPIES)



Telomeres maintain the stability of chromosomes, ensure proper replication of DNA ends, and serve as a sign for chromosome health. Often shortened in each replication cycle and maintained in various tumors via telomerase.



AMOUNT OF DNA DOUBLES
OF CHROMOSOMES DOES NOT ↑

↳ HUMAN SOMATIC CELLS = 46 CHROMOSOMES

↳ AMOUNT OF DNA CHANGES DURING CYCLE

G2 PHASE ~ GAP OR GROWTH 2

CELL MUST DUPLICATE ORGANELLES

CYTOPLASM ORGANELLES



(END OF G2)

FINAL G2 CHECKPOINT

*NO DNA DAMAGE
AFTER REPLICATION



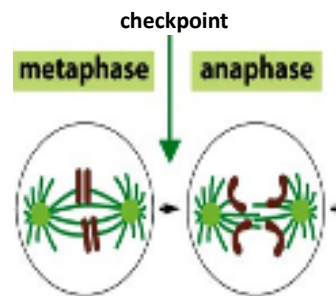
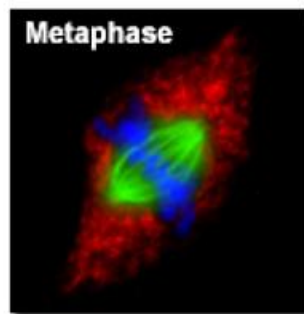
ENTER MITOSIS

OSMOSIS.org

Mitosis

(Karyokinesis and end w. cytokinesis)

The phases of mitosis in epithelial kidney cells. Microtubules stained green, DNA stained blue and mitochondria: stained red.



[Molecular Expressions: Images from the Microscope \(fsu.edu\)](#)

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A) Prophase

Nuclear membrane disintegrate, chromatin is organized into chromosomes. The two-centriole pairs migrate to opposite poles of the cell forming the centrosome, cytoplasmic microtubules polymerize to prepare for the formation of mitotic spindle fibers

b) Metaphase

Chromosomes line up at the cell equator & microtubules form the mitotic spindles.

Inhibitor of sister chromatid separation (ISS) protein holds the two sister chromatids together until all chromosomes are properly aligned at the metaphase plate and degraded afterwards.

c) Anaphase

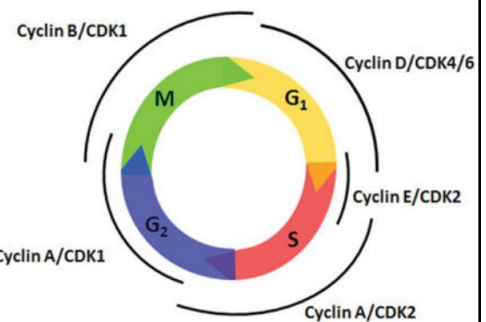
chromatids are pulled apart by the spindle apparatus and migrate to the opposite sides of the cell.

D) Telophase: Final stage. Daughter chromosomes are at the spindle poles. The nuclear membranes are formed at this stage.

Cytokinesis: the process where the cytoplasm is divided by cleavage. It starts in the late anaphase and continues through telophase.

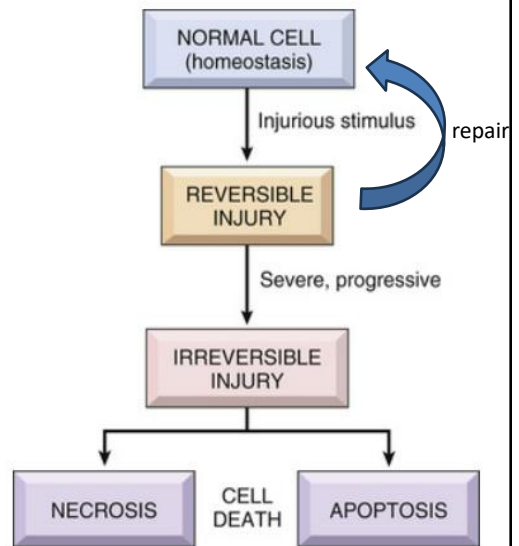
Cell Cycle Control

- Different checkpoints serve as control system that respond to feedback (malfunction in DNA synthesis delay progression to the M phase).
- Families of regulatory molecules, among them:
 - Cyclins & CDKs (cyclin-dependent kinases)
 - Different cyclins activate their respective (CDK) to determine cell cycle progression (example: activation of S-phase cyclin-CDK complexes (s-CDKs) initiates the S phase).
 - Cyclin-dependent kinase inhibitors (CKI) can stop or arrest the cell cycle at various phases.



Cell Injury

- 1- Hypoxia/ischemia injuries
- 2- Toxins (CO, drug toxicity).
- 3- Pathogenic infections can cause cell injury by various mechanisms.
- 4- Immune defects, autoimmune, allergic or hypersensitivity reactions.
- 5- Genetic defects.
- 6- Nutritional deficiencies
- 7- Physical damage like in cases of trauma, extremes temperature fluctuations, radiation.
- 8- Aging decrease ability of cells to respond to stress.
- 9- Variable stressors leading to cellular adaptations.



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Cell Death

APOPTOSIS

* PROGRAMMED DEATH

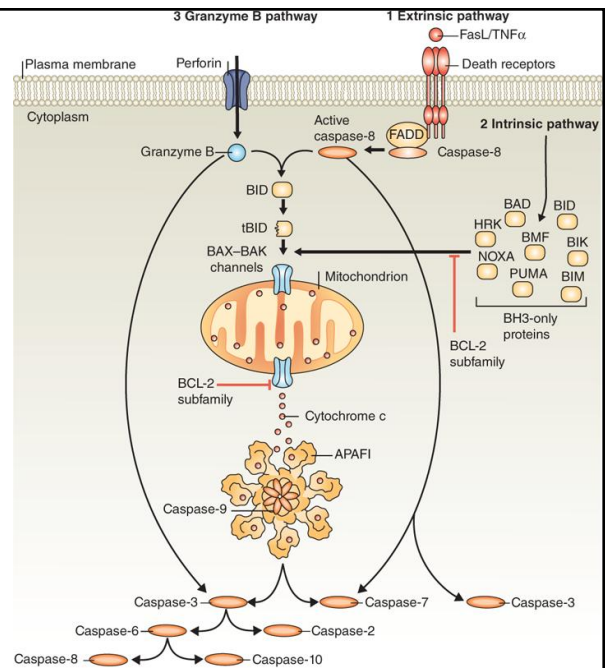
NECROSIS

* INJURY

* DISEASE

Apoptosis (programmed Cell Death)

- Active process of cellular self-destruction that occurs in normal and pathologic conditions.
- Needs ATP for energy.
- Can be intrinsic or extrinsic



Source: Howard M. Reisner
Pathology: A Modern Case Study, Second Edition
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Intrinsic Pathway

(aka, mitochondrial pathway).

- Exposure to oxidative stress, hypoxia, protein misfolding.
- Activates proapoptotic signals leading to activation of caspase signaling (procaspase to caspase)

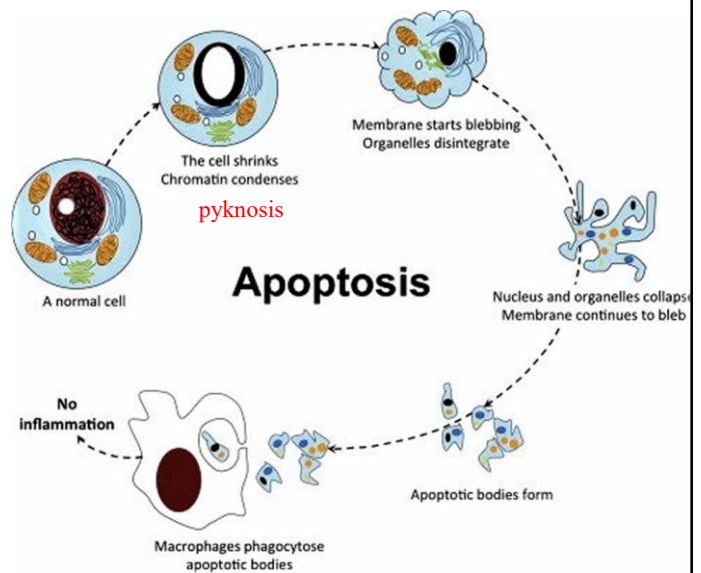
Extrinsic pathway

(aka, death receptor pathway)

- Exposure to death signals from outside cell (e.g: $\text{TNF}\alpha$) will activate a sequence of molecular signaling events that finally activate caspase signaling.

Caspase signaling cause cleavage of cellular proteins and the cytoskeleton. The weakened cell membrane develop blebs that can break off and phagocytosed by macrophages. Intactness of the membrane in apoptosis reduces inflammation.

- ❖ Membranes remain intact.
- ❖ Cell size reduces
- ❖ No Inflammation.
- ❖ Failure may lead to tumor development
- ❖ Regulated pathway:
e.g overexpression of BCL-2 “anti-apoptosis” leads to follicular lymphoma)



[Apoptosis definition, cell apoptosis pathway, steps & apoptosis inducer \(healthjade.net\)](#)

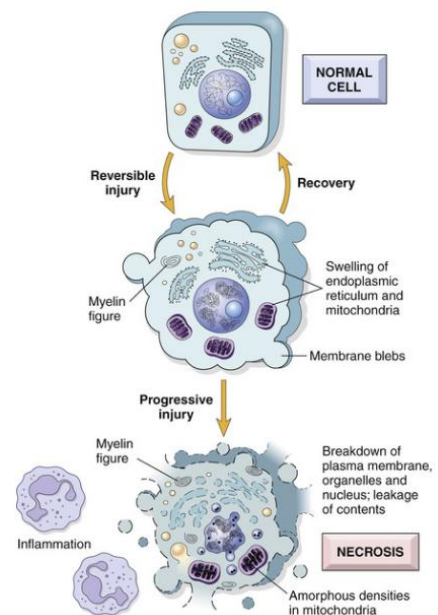
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Necrosis

Necrosis can occur secondary to internal factors (like ischemia) or external factors (like infections or extreme temperature fluctuations).

Cells swell and burst which triggers an inflammatory response that can cause further membrane damage. Nucleus undergo pyknosis (shrinkage with dense genetic material) followed by karyolysis (nuclear dissolution and lysis of chromatin).



FEATURES OF NECROSIS AND APOPTOSIS

FEATURE	NECROSIS	APOPTOSIS
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-size fragments
Plasma membrane	Disrupted	Intact; altered structure, especially orientation of lipids
Cellular contents	Enzymatic digestion; may leak out of cell	Intact; may be released in apoptotic bodies
Adjacent inflammation	Frequent	No
Physiologic or pathologic role	Invariably pathologic (culmination of irreversible cell injury)	Often physiologic, means of eliminating unwanted cells; may be pathologic after some forms of cell injury, especially DNA damage

From Kumar V et al: Cellular responses to stress and toxic insults: adaptation, injury, and death. In Kumar V, Abbas A, Fausto N: *Robbins & Cotran pathologic basis of disease*, ed 8, St Louis, 2010, Saunders.

[VitalSource Bookshelf: Pathophysiology](#)

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Types of necrosis

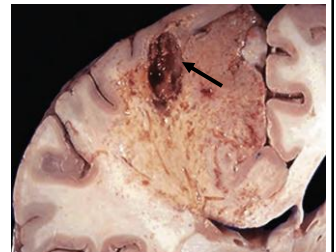
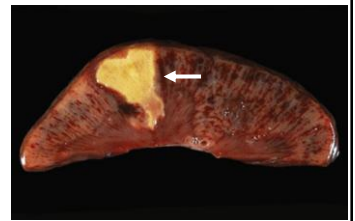
Coagulative necrosis

Secondary to ischemic injury, observed in multiple organs like spleen, kidney, heart and liver.

Caused by protein denaturation (inactivation of lysosomal enzymes) leading to a coagulative, gelatinous, opaque-clear appearance with preserved tissue structure that stays intact for days.

Liquefactive necrosis

In cases of ischemic injury to the brain, and bacterial abscess. The release of hydrolytic enzymes will lead complete digestion of damaged area leading to soft, liquefied tissue appearance.



Gangrenous necrosis

Secondary to ischemia. Can be observed in lower limbs or GI.

Dry gangrene has coagulative necrosis microscopic appearance, tissue appear dried, no infection and has brownish black color.

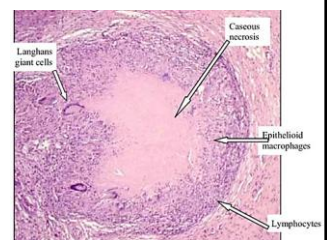
Wet gangrene has liquefactive necrosis appearance due to bacterial and inflammatory leukocytes action.



Caseous necrosis

Usually observed in lungs (secondary to tuberculosis).

Morphologically, the affected area will have a soft, granular, yellowish “cheese-like” appearance. The necrotic focus will be surrounded by macrophages and other inflammatory cells yielding granulomatous inflammation to isolate this area.



[\(Cytokines and tuberculosis : an investigation of tuberculous lung tissue and a comparison with sarcoidosis \(researchgate.net\)\).](#)

[Gangrene | The Atlas of Emergency Medicine, 5e | AccessMedicine | McGraw Hill Medical \(cnsu.edu\)](#)

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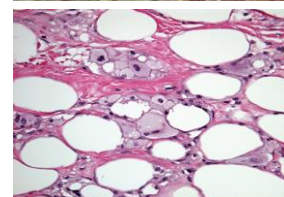
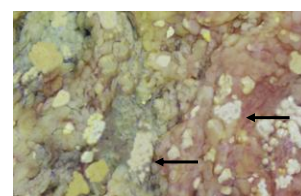
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Fat necrosis

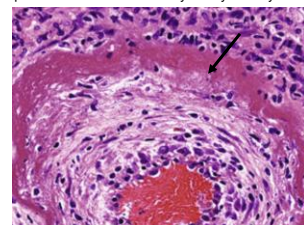
Can be triggered by trauma to the breast, acute pancreatitis. Caused by leakage and digestive action of lipases on adipocytes. Lipases break down triglycerides (yielding free fatty acids) that undergo saponification process when combined with calcium. Morphologically, affected area will have chalky white-like appearance.

Fibrinoid necrosis

For blood vessels. Appear under the microscope as necrotic connective tissue that histologically resembles deposited fibrin fibers. Occur due to hypersensitivity reactions (II and III) and blood vessel damage in hypertensive crisis.



Source: Howard H. Rensier
Pathology: A Modern Case Study, Second Edition
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Fat necrosis. The adipose tissue contains multiple clear cystic spaces that are surrounded by foamy histiocytes.



**Failure of cell cycle regulation or programmed
cell death may lead to cancer**

What is Cancer?

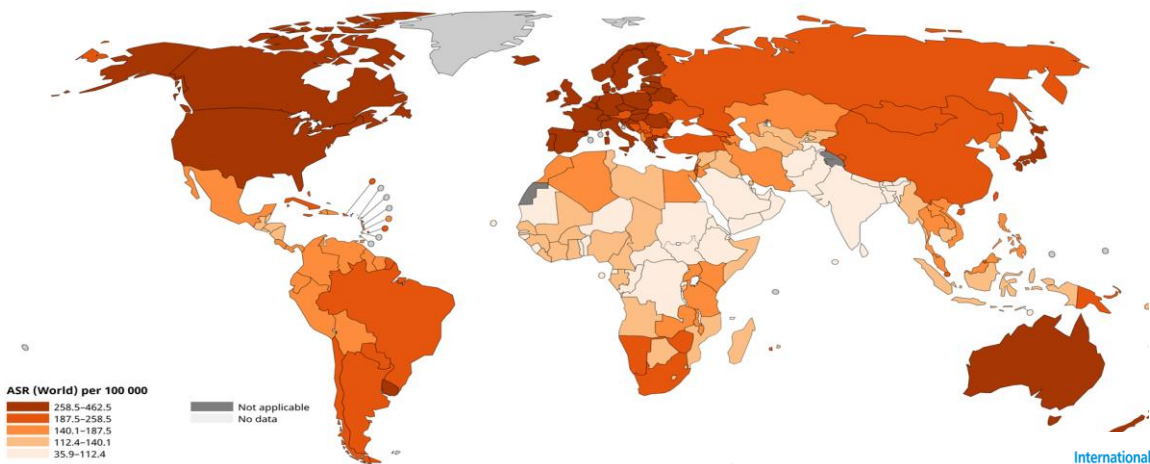
- Cancer can be defined as “diseases in which abnormal cells divide without control and are able to invade other tissues.”
- Neoplasia = “new growth” Neoplasm refers to new and abnormal tissue growth.
 - Benign tumors remain localized and can undergo surgical removal.
 - Malignant tumor refers to lesions that can invade adjacent tissues and metastasize. Malignant tumors are referred to as cancer.

Tumors have two basic components:

- 1- Parenchyma**, made up of transformed or neoplastic cells.
- 2- Non-neoplastic stroma:** constitute supporting host-derived tissue made up of connective tissue, blood vessels, and host-derived inflammatory cells. Constitute the tumor microenvironment.

Global Cancer Burden

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
All cancers



Cancer TODAY | IARC
<https://gco.iarc.who.int/today>
 Data version: Globocan 2022 (version 1.1) - 08.02.2024
 © All Rights Reserved 2025

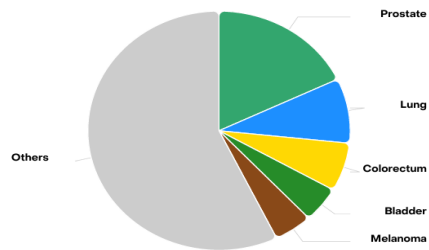
International Agency
for Research on Cancer
 World Health Organization

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Top 5 most frequent cancers**

Males

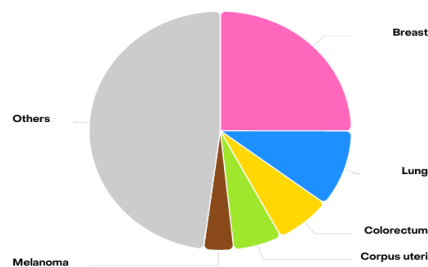


Total: 1 283 898

Rank	Cancer site	Number of cases	Percent
1st	Prostate	230 125	17.9%
2nd	Lung	112 343	8.8%
3rd	Colorectum	84 172	6.6%
4th	Bladder	61 657	4.8%
5th	Melanoma	60 762	4.7%
-	Others	734 839	57.2%

Number of new cases in 2022, males, all ages

Females



Total: 1 096 291

Rank	Cancer site	Number of cases	Percent
1st	Breast	274 375	25.0%
2nd	Lung	113 690	10.4%
3rd	Colorectum	76 014	6.9%
4th	Corpus uteri	66 055	6.0%
5th	Melanoma	40 626	3.7%
-	Others	525 531	47.9%

Number of new cases in 2022, females, all ages

International Agency
for Research on Cancer



World Health
Organization

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Causes of Cancer

A) Genetic risk factors:

Numerous gene mutations (acquired and germline) can increase risk of cancer via multiple mechanisms

For example, germline mutations include:

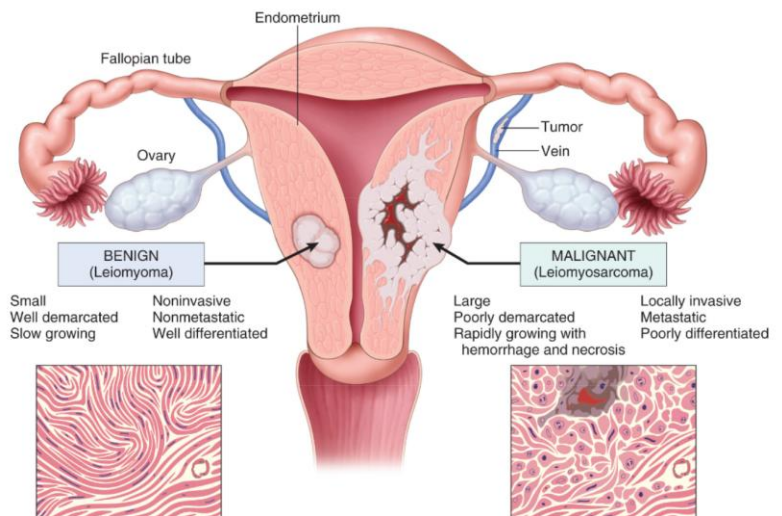
- *BRCA-1 and -2* mutations increases risk for breast cancer and ovarian cancer.
- *APC* mutations causes Familial Adenomatous Polyposis (FAP) which increases the patient risk for several cancers including for colon and rectal cancer.

B) Environmental risk factor:

- Mutagenic chemical agents (alkylating agents, metal ions,...)
 - Infection with oncogenic viruses (e.g., human papilloma virus types “HPV” (↑risk of cervical/oropharyngeal cancer); Epstein-Barr virus “EBV” (↑risk of B-cell lymphoma). Hep. B and Hep. C infection (↑risk of hepatocellular carcinoma).
 - Ionizing and ultraviolet radiation can cause DNA damage and increase the risk of cancer (e.g., UV & skin cancer).
 - Smoking: associated with several types (e.g., lung cancer)
 - Others: diabetes, obesity, alcohol, sedentary lifestyle.
- C) others:** cell cycle dysregulation and genome instability.....

Benign vs Malignant tumors

- Degree of differentiation.
- Rate of growth.
- Local invasiveness.
- Distant spread.



Comparison between a benign tumor of the myometrium (leiomyoma) and a malignant tumor of similar origin (leiomyosarcoma).

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Nomenclature

- Benign tumors of epithelia or mesenchyme often ends with ---OMA. Malignant tumors:--carcinoma (epithelial origin) or –sarcoma (mesenchymal origin).

Origin	Benign	Malignant
Epithelium	Adenoma (benign neoplasm of glandular epithelium.	Adenocarcinoma (malignant neoplasm of glandular tissue).
Mesenchyme	osteoma and chondroma (benign bone and cartilage tumor, respectively).	Rhabdomyosarcoma (malignant neoplasm of skeletal muscle Osteosarcoma (malignant neoplasm of bone).

Cancer Genes

Inherited Predisposition	Gene(s)
Autosomal Dominant Cancer Syndromes	
Retinoblastoma	<i>RB</i>
Li-Fraumeni syndrome (various tumors)	<i>TP53</i>
Melanoma	<i>CDKN2A</i>
Familial adenomatous polyposis/colon cancer	<i>APC</i>
Neurofibromatosis 1 and 2	<i>NF1, NF2</i>
Breast and ovarian tumors	<i>BRCA1, BRCA2</i>
Multiple endocrine neoplasia 1 and 2	<i>MEN1, RET</i>
Hereditary nonpolyposis colon cancer	<i>MSH2, MLH1, MSH6</i>
Nevoid basal cell carcinoma syndrome	<i>PTCH1</i>
Autosomal Recessive Syndromes of Defective DNA Repair	
Xeroderma pigmentosum	Diverse genes involved in nucleotide excision repair
Ataxia-telangiectasia	<i>ATM</i>
Bloom syndrome	<i>BLM</i>
Fanconi anemia	Diverse genes involved in repair of DNA cross-links

“Figure for Illustration”

A) Oncogenes

- Mutated form of proto-oncogenes.

A proto-oncogene is a gene that normally stimulates cell growth and division.

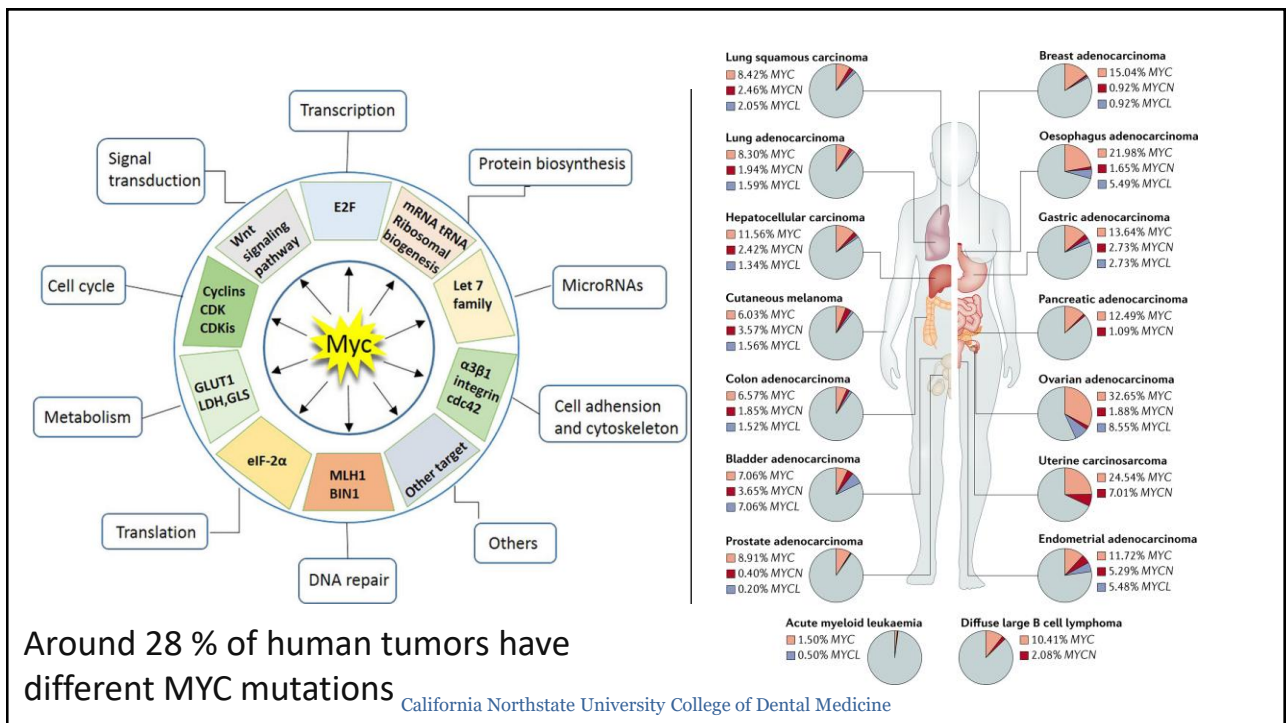
- Most oncogenes encode for transcription or growth factors to enhance cell survival. A single allele mutation is usually sufficient to produce an oncogenic effect.
- Oncogenes can induce cell transformation:
 - *K-RAS* mutations are observed in pancreatic, lung, colon, and endometrial tumors.
 - *N-RAS* mutations are observed in melanomas.
 - *H-RAS* mutations are observed in cancers of the adrenal gland and thymus.

ONCOGENES

- * **MUTATED** VERSIONS of
PROTO-ONCOGENES
↳ **POSITIVE REGULATION** of CELL CYCLE
- * **PROTEIN PRODUCTS STIMULATE**
CELL GROWTH & DIVISION



GAS PEDAL



B) Tumor suppressor genes

- Normally prevent uncontrolled cell growth.
- Upon the mutation or loss of function of those genes, the transformed cell phenotype can develop.
- In most cases, functional loss of both tumor suppressor gene alleles is required for cell transformation. For example, loss of function mutation in *P53* genes.

TUMOR SUPPRESSOR GENES

* **NEGATIVE REGULATION of CELL CYCLE**

* **PROTEIN PRODUCTS STOP CELL CYCLE PROGRESSION & PROMOTE APOPTOSIS**

↳ INVOLVED in DNA REPAIR

↳ INHIBIT TRANSCRIPTION FACTORS that PUSH ALONG CELL CYCLE

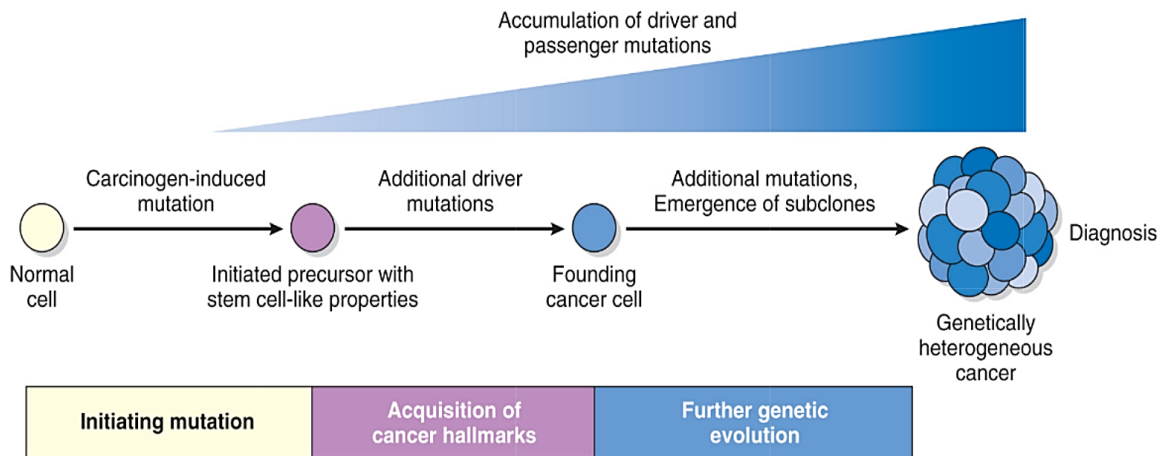


BRAKE PEDAL  **2022 Edition**

C) Genes that regulate apoptosis

- For example, BCL-2 family of proteins
 - Some members (e.g: *BAX*) are pro-apoptotic. Activated following cell damage or normal cell turnover mechanisms. In cancer cells, those genes are often down-regulated or functionally inactivated by mutations.
 - Other members are anti-apoptotic (*BCL-2*). Upregulated expression of BCL2 inhibits apoptosis, seen in certain tumors like multiple myeloma.

Carcinogenesis



Tumor progression continued evolution and fitness to the selective pressure of immune system and drug therapy.

Driver mutations (also named pathogenic mutations)

- They directly contribute to the development or progression of cancer (e.g oncogenes, tumor suppressor genes).
- These are mutations that alter the gene function and thus give a growth advantage to the prospective tumor cell.
- They can be acquired or inherited.

Passenger mutations (also named neutral or hitchhiker mutations).

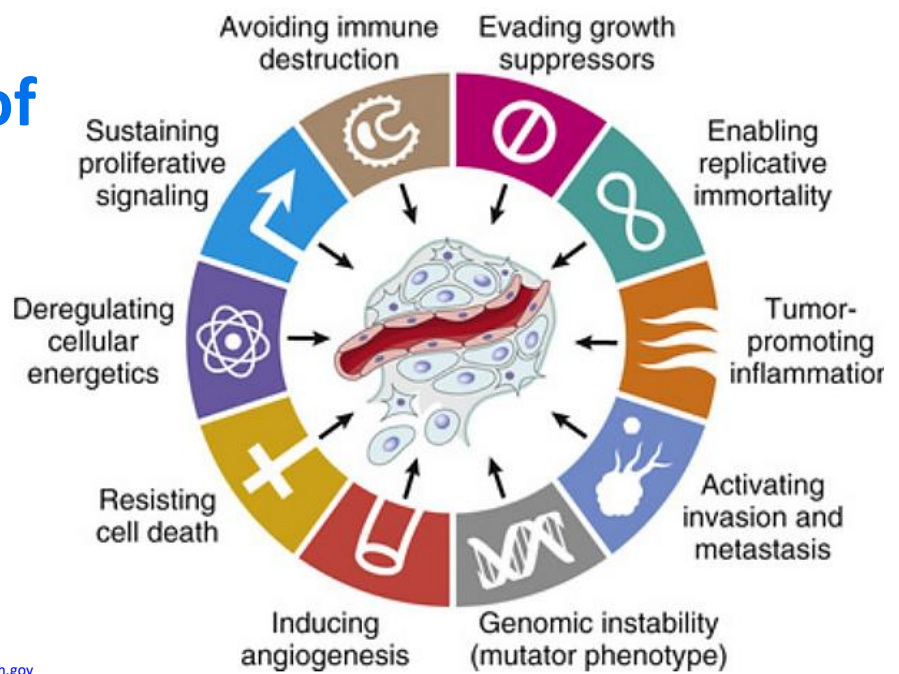
- They are acquired mutations that do not normally affect the cellular growth behavior.
- They do not initiate carcinogenesis, but they aid the cancer formation process.
- Passenger mutations can increase the risk of drug resistance.

In a tumor, the mixture of passenger and driver mutations together comprise the mutated gene sets (MGS) of that tumor.

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Hallmarks of Cancer



[The hallmarks of cancer - PubMed \(nih.gov\)](#)

[Hallmarks of cancer: the next generation - PubMed \(nih.gov\)](#)

[Revisiting the hallmarks of cancer - PMC \(nih.gov\)](#)

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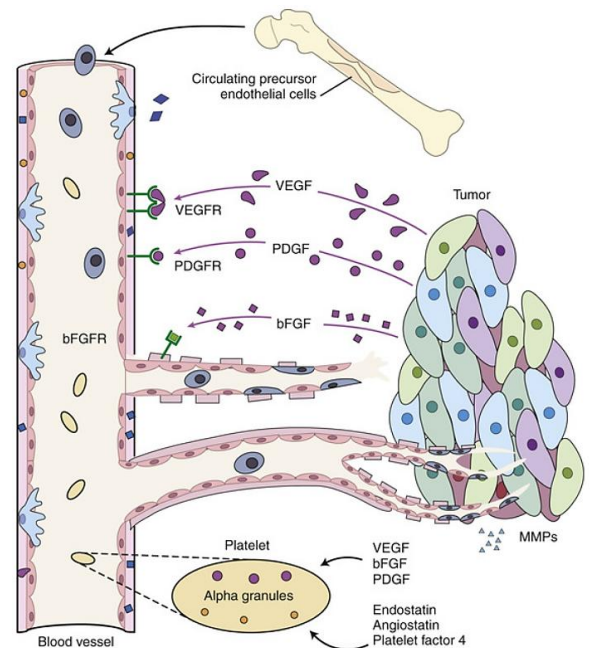
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Sustained angiogenesis

Angiogenic factors and angiogenic inhibitors control the development of new vessels.

To secure enough nutrients, tumors can manipulate the surrounding environment to induce angiogenesis.

Cancer cells use several mechanisms to increase the secretion of angiogenic factors and decrease the release of angiogenic inhibitors.

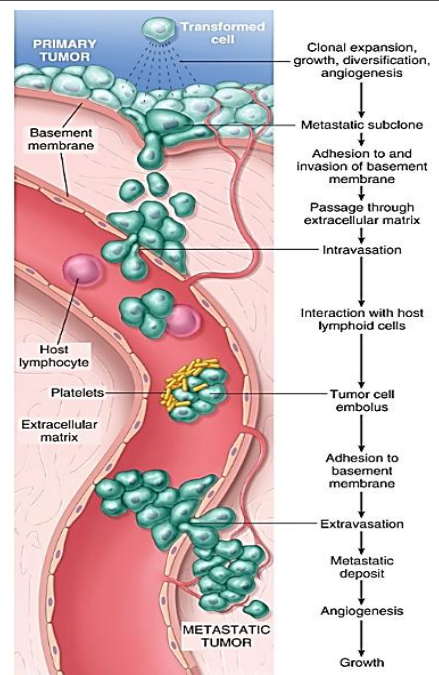


Invasion and metastasis

Cancer metastasis is a major cause of cancer-related morbidity and mortality.

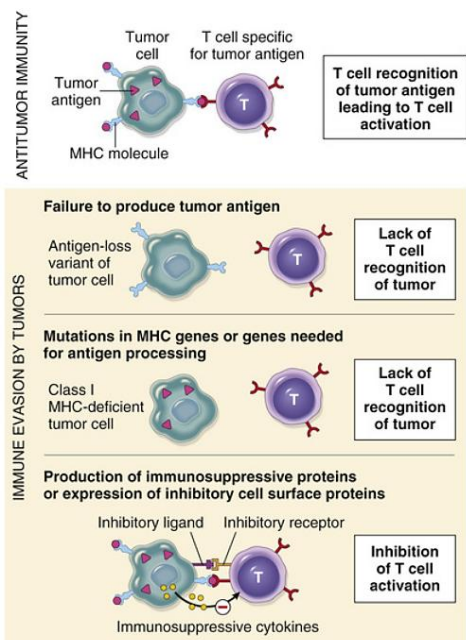
It can occur via:

- 1- Loosening of existent cell–cell contacts through the inactivation of E-cadherin.
- 2- Degradation of extracellular matrix.
- 3- Migration and invasion



Evade immune destruction

- Tumors express “tumor associated antigens” and “tumor specific antigens”.
- Successful immunosurveillance response can lead to the spontaneous regression of some tumors.
- Tumors can induce immune evasion by multiple mechanisms.



[VitalSource Bookshelf: Pathophysiology](#) California Northstate University College of Dental Medicine

Deregulating cellular energetics

Cancer cells have characteristic form of cellular metabolism (**Warburg effect**).

- Cancer cells consume large amount of glucose to generate energy quickly and utilize the generated lactate as an amino acid precursor to support the growth of daughter tumor cells.

(Normal cell metabolism converts glucose to pyruvate and proceed to Krebs cycle to maximize glucose use).

Manifestations, diagnosis and management strategies

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Systemic effects of Neoplasia

1- **Effect of the tumor mass:** depend on the tumor size and location. Include physical pressure or loss of normal tissue function, bone marrow failure, ischemia, ...

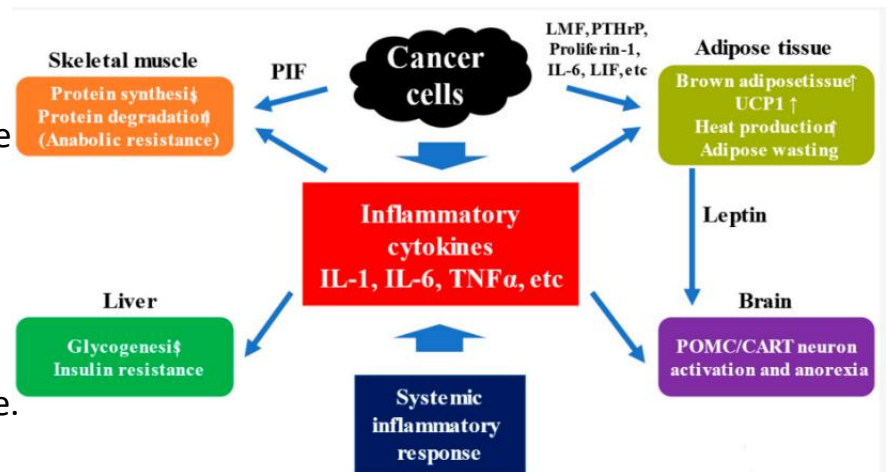
2- **Paraneoplastic Syndromes.**

Symptoms triggered by the tumor but not attributed to the direct effect of the tumor mass. Caused by

- a) Substances released from the tumor (for example: hormones).
- b) Immune response targeting the tumor.

3- Cancer-induced Cachexia

Cachexia leads to excessive weight loss, marked weakness, taste alterations, muscle wasting, as well as altered protein, lipid, and carbohydrate metabolism and decreased quality of life.



[Cancer Cachexia: Its Mechanism and Clinical Significance - PubMed \(nih.gov\)](#)

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NOT for Illustration"

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Pathogenesis of cachexia, may be related to mediators released by cancer cells like:

- 1- Proteolysis-inducing factor (PIF) induces protein degradation, apoptosis via activation of caspases, inhibits protein synthesis, increase proteolysis
- 2- Lipid-mobilizing factor (LMF): promotes lipolysis of adipose tissue and increase fatty acid oxidation.
- 2- Parathyroid hormone-related protein (PTHrP) causes calcium mobilization from bone and hypercalcemia.
- 3- Proinflammatory cytokines (TNF alpha, interferon gamma and IL-6 are central) contribute to inflammatory process and generalized weakness.

Diagnosis

Tumor diagnosis and staging uses a combination of:

- Clinical exam
- Imaging
- Tissue biopsies and histochemistry/immunohistochemistry
- Blood biochemistry and tumor markers
- Genetics / immunophenotyping

Tumor markers

- Tumor markers include hormones, enzymes, genes, antigens, and antibodies.
 - Alpha fetoprotein is elevated in hepatocellular carcinoma.
 - CD20 is elevated in non-Hodgkin's lymphoma.
 - Prostate-specific antigen (PSA) increases in prostate cancers
 - Benign tumors of the adrenal medulla (pheochromocytoma) can produce catecholamines (for example: adrenaline) in large quantities
- Tumor markers can be used for:
 - (1) To identify individuals at high risk for cancer.
 - (2) To aid in the diagnosis of a specific type of tumor (ACTH for pituitary tumors, AFP can indicate liver cancer). Level is not prognostic for survival.
 - (3) To follow the clinical course of treatment and monitor remission.

Management Strategies

- Surgery
- Radiation therapy
- Chemotherapy
- Immunotherapy
- Targeted therapy

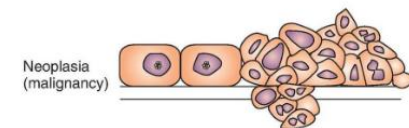
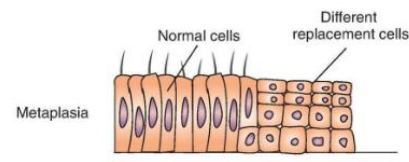
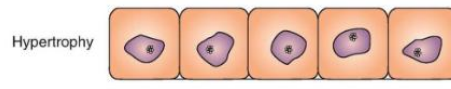
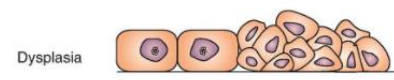
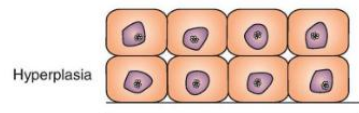
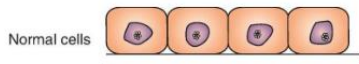




[Cancer Prevention Recommendations - WCRF International](https://www.wcrf.org/cancer-prevention-recommendations)

California Northstate University College of Dental Medicine

Cellular Adaptation Terminology



Atrophy: Decrease in cell size (yielding reduced tissue mass)

Hypertrophy: Increase in individual cell size (muscle hypertrophy)

Hyperplasia: the cells increase in number in an organ.

Dysplasia: development of immature cells with abnormal shapes, that is not typical for healthy tissue (pre-cancerous).

Metaplasia: reversible change in cell type to adapt to stress (one differentiated cell type is replaced by another differentiated cell).

Neoplasia “new growth”. Neoplasm refers to new and abnormal tissue growth.

- **Anaplasia** refers to a lack of differentiation in neoplastic cells. Well-differentiated tumors resemble their tissue of origin. Anaplastic tumor cells are poorly-differentiated (don't resemble tissue of origin, poor prognosis).

Aplasia: complete congenital lack of a tissue.

EVERY DAY, MILLIONS OF LIVES ARE
TURNED UPSIDE DOWN BY CANCER

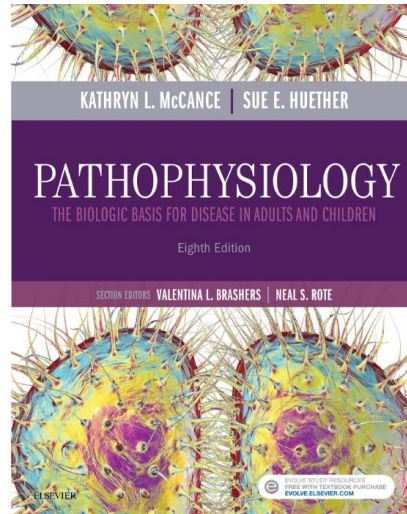
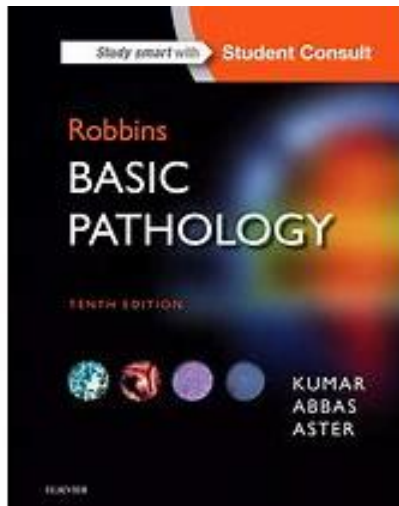
#LivesTurnedUpsideDown



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Resources



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